A-LEVEL BIOLOGY SEMINAR
HOLY CROSS LAKE VIEW
Prepared by
Association of Biology Educators

The Association of Biology Educators warmly welcomes you to this A-Level Biology seminar.

As you prepare for your final examinations, remember that success in Biology requires more than recalling facts. You must apply biological knowledge and practical skills to unfamiliar situations.

During this seminar, you will be guided on how to analyse and evaluate scenario-based items by interpreting data, identifying trends, using evidence, explaining biological mechanisms and drawing logical conclusions. You will also learn to evaluate biological responses and interventions by stating whether they are **effective or not effective,** explaining their mechanisms and effectiveness, identifying limitations and making appropriate recommendations.

Special attention will also be given to **practical investigations, microscopy and dissection**. You will learn how to plan investigations, identify variables, record and interpret results, use a microscope correctly, prepare and observe specimens, make accurate biological drawings, and dissect specimens safely to identify important structures.

Participate actively, ask questions and practice every task carefully. Use this opportunity to strengthen your knowledge, practical skills, confidence and examination technique.

On behalf of the Association of Biology Educators, I wish you a productive seminar and success in your final examinations.

Thank you.

CONSTRUCT 1

ITEM 1

Medical officers investigated workers who had spent several years in a chemical-processing factory. Some workers had been exposed to a substance that altered a tumour-suppressor gene. Another chemical affected the inner mitochondrial membranes of lung cells. During sample preparation, chemical residues entered some DNA samples and interfered with DNA polymerase.

The affected workers developed lung tumours and could no longer perform strenuous physica work. Confirmation of the genetic condition took longer than expected, delaying the selection c suitable treatment. The laboratory findings are shown in Table.

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Table: Findings from the affected workers

Parameter	Normal finding	Finding in affected workers
Cells completing mitosis per hour	34	182
ATP concentration in lung tissue (mmol kg^{-1})	5.3	1.9
Copies of the target DNA sequence after PCR	860	36

Health officers proposed the following interventions:

- administer a drug that prevents rapidly dividing cells from completing mitosis;
- provide concentrated lipid supplements to increase energy production;
- purify the DNA samples before repeating PCR.

Task

(a) Analyse how the changes shown in Table contributed to tumour development, affected ability to perform strenuous work and selection of treatment.

(b) Assess the effectiveness of the proposed interventions in controlling tumour development, improving the workers' physical activity and obtaining a reliable genetic diagnosis.

ITEM 2

A metal-processing factory in Jinja employed workers from different family backgrounds. One worker, Musa, had inherited an altered gene for the beta-globin chain of haemoglobin. During an industrial accident, he inhaled cyanide fumes. A solvent also entered his bloodstream and affected glucose-carrier proteins in the plasma membranes of muscle cells.

Several hours later, Musa developed severe joint pain, repeated muscle cramps and early muscle fatigue. The findings from his medical examination are shown in the table below.

Table: Musa's laboratory findings

Parameter	Musa's result	Normal value
Abnormally shaped red blood cells (%)	62	Below 5
Glucose uptake by muscle cells (arbitrary units)	38	90
ATP concentration in muscle cells (mmol kg^{-1})	1.8	5.2

Medical workers used the following interventions:

- transfusion with normal red blood cells;
- administration of hydroxocobalamin as an antidote to cyanide;
- intravenous administration of a concentrated glucose solution.

Task

(a) (i) Analyse the relationship between the inherited genetic change and Musa's joint pain.

(ii) Explain how the changes in membrane transport and cellular respiration contributed to the muscle cramps and early muscle fatigue.

(b) Evaluate the effectiveness of each medical intervention in improving oxygen delivery and energy production in Musa's tissues.

ITEM 3

A metal-processing factory in Jinja employed workers from different family backgrounds. One worker, Musa, had inherited an altered gene for the beta-globin chain of haemoglobin. During an industrial accident, he inhaled cyanide fumes. A solvent also entered his bloodstream and affected glucose-carrier proteins in the plasma membranes of muscle cells.

Several hours later, Musa developed severe pain in the joints, repeated muscle cramps and early muscle fatigue. The findings from his medical examination are shown in Table.

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Table: Musa's laboratory findings

Parameter	Musa's result	Normal value
Abnormally shaped red blood cells (%)	62	Below 5
Glucose uptake by muscle cells (arbitrary units)	38	90
ATP concentration in muscle cells (mmol kg^{-1})	1.8	5.2

Musa's cells showed the following responses:

- increased production of red blood cells;
- increased anaerobic breakdown of glucose;
- increased activity of enzymes of the Krebs cycle.

Task

(a) (i) Analyse the relationship between the inherited genetic change and Musa's joint pain.

(ii) Explain how the changes in membrane transport and cellular respiration contributed to the muscle cramps and early muscle fatigue.

(b) Assess the effectiveness of the cellular responses in improving oxygen delivery and energy production in Musa's tissues.

CONSTRUCT 2

ITEM 1

Coffee farmers on the slopes of Mount Elgon experienced an extended dry season. Closely spaced shade trees reduced the light reaching the coffee leaves. The farmers had also planted seeds that remained dormant during storage, while intact shoot apices restricted the development of lateral branches.

The coffee gardens later produced few seedlings, poor branching, flower loss and low berry yields. An agricultural officer obtained the findings shown in Table 4.

Table : Findings from the coffee gardens

Parameter	Well-managed garden	Affected garden
Stomatal conductance (arbitrary units)	74	28
Rate of non-cyclic photophosphorylation (relative units)	82	34
Active gibberellin concentration in seeds (relative units)	7.8	2.3
Lateral branches per mature plant	24	9

The following agricultural interventions were proposed:

- apply organic mulch around the coffee plants;
- remove selected branches from the shade trees;
- soak the seeds in an appropriate gibberellin solution before planting;

Task

(a) (i) Analyse how the changes in stomatal conductance and non-cyclic photophosphorylation contributed to flower loss and low berry yield.

(ii) Explain how changes in gibberellin concentration and lateral-branch development affected seedling establishment and branching.

(b) (i) Determine the effectiveness of mulching and shade management in improving coffee productivity.

(ii) Evaluate the effectiveness of gibberellin treatment and auxin application in improving seedling establishment and branching.

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ITEM 2

Farmers in Isingiro District planted bean seeds that had been stored from the previous growing season. During germination and early growth, the area experienced prolonged drought and high temperatures. Some of the stored seeds had hard seed coats.

The farmers observed poor crop establishment, wilting, slow accumulation of dry matter and reduced primary growth of the young plants. Field investigations produced the findings shown in Table 1.

Table 1: Findings from the bean plants

Parameter	Plants under suitable conditions	Affected plants
Carbon dioxide uptake ($\mu mol\ m^{-2}\ s^{-1}$)	18.4	6.2
Soil water available to roots (%)	76	23
Seeds completing germination (%)	91	37
Increase in shoot length per week (cm)	8.2	2.6

Some of the affected plants showed the following responses:

- increased secretion of abscisic acid;
- development of an extensive root system;

Task

(a) (i) Account for the wilting and slow accumulation of dry matter using the soil water available to roots and carbon dioxide uptake shown in Table 1.

(ii) Analyse how the percentage of seeds completing germination and the increase in shoot length influenced crop establishment and primary growth.

(b) Assess the effectiveness of the observed responses in promoting the survival and productivity of the bean plants under the prevailing conditions.

ITEM 3

Farmers compared the productivity of a short-day C_3 soybean variety grown in the lowlands of Kasese and the highlands of Kabale. The Kasese crop experienced higher daytime temperatures than the Kabale crop. Both crops were planted during a season when daylight lasted longer than that required by the variety. Some Kasese gardens also remained waterlogged after heavy rainfall.

The lowland crop accumulated little dry matter, both crops flowered late, and many pods from the waterlogged gardens fell before maturity. The investigation findings are shown in Table below.

Table: Findings from soybean plants in the two regions

Parameter	Kabale highlands	Kasese lowlands
Photorespiration rate (relative units)	18	52
Florigen concentration at the shoot apex (relative units)	4.2	3.5
Ethylene concentration in pods (relative units)	1.3	4.9
Dry mass accumulated per plant (g)	46	24

Agricultural officers proposed the following interventions:

Task

(a) (i) Compare the performance of the C3 soybean plants in the highland and lowland areas.

(ii) Explain how photoperiod and ethylene activity affected flowering and fall of pods.

(b) Propose the most effective agricultural practices to solve the challenges above.

One strength explained
One limitation explained
Recommendation
Mitigation (to the limitation)

S - Insulin stimulates conversion of excess glucose to glycogen for storage in liver cells

4 - Excess insulin in the body results into hypoglycaemia (low levels of glucose)

R -

Mitigation;

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ITEM 4

Agricultural researchers grew equal numbers of bean plants in two different greenhouses. Greenhouse A was well ventilated and received controlled irrigation. Greenhouse B was poorly ventilated and excessively irrigated, causing high humidity and prolonged waterlogging of the soil. Both greenhouses received equal light intensity and contained the same soil type.

Plants in Greenhouse B developed yellow leaves, showed poor growth and produced fewer pods. Their flowers also remained wet for long periods, reducing successful pollination. The investigation findings are shown in the table below.

Table: Findings from bean plants grown in the two greenhouses

Parameter	Greenhouse A	Greenhouse B
Nitrate-ion concentration in leaves (mg g^{-1})	7.8	3.1
Flowers successfully pollinated (%)	84	39
Mean number of pods produced per plant	26	11
Mean mass of seeds produced per plant (g)	48	19

Task

(a) (i) Analyse the performance of the bean plants grown in Greenhouse A and Greenhouse B.

(ii) Explain how excessive irrigation and high humidity affected the growth and reproduction of the plants in Greenhouse B.

(b) Propose and evaluate the most effective agricultural practices for improving the growth and yield of the bean plants in Greenhouse B.

SECTION B
PART I

CONSTRUCT 3

ITEM 1

Firefighters rescued people from a burning building in Jinja City. One firefighter remained in dense smoke for several minutes. Immediately afterwards, he entered a dark room after looking directly at bright flames. His protective clothing also prevented effective loss of body heat.

The firefighter became confused, failed to identify objects in the dark room and developed heat exhaustion. Medical findings recorded at different times are shown in Table.

Table: Findings from the firefighter after the rescue

Time after rescue	Carboxyhaemoglobin (%)	Rhodopsin remaining in rod cells (%)	Body temperature (°C)
Immediately	32	18	40.3
After 30 minutes	26	46	39.7
After 60 minutes	19	73	38.9
Normal condition	Below 2	Above 90 in darkness	37.0

Medical officers proposed the following interventions:

- administer oxygen at a high concentration;
- place the firefighter under bright light before he returns to the dark room;
- remove the heavy protective clothing and cool his body with water.

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Task

(a) (i) Analyse how the changes in carboxyhaemoglobin and body temperature contributed to confusion and heat exhaustion.

(ii) Explain how the change in rhodopsin affected vision in the dark room.

(b) Evaluate the effectiveness of the proposed interventions in restoring oxygen transport, dark vision and normal body temperature.

ITEM 2

Workers at a food-storage facility in Kampala operated a charcoal stove in a poorly ventilated room. One of the workers had spent several hours in cold weather before entering the room. The workers had also missed their morning meal because they reported for work early.

After some time, one worker developed a severe headache, became physically weak and responded slowly when spoken to. Medical investigations produced the findings shown in Table 1.

Table 1: Medical findings from the affected worker

Parameter	Normal value	Worker's value
Haemoglobin combined with carbon monoxide (%)	Below 2	38
Speed of nerve impulse transmission ($m\ s^{-1}$)	84	41
Blood glucose concentration ($mmol\ L^{-1}$)	5.0	2.8

The worker's body showed the following physiological responses:

- increased breathing rate;
- shivering;
- increased secretion of glucagon.

Task

(a) (i) Analyse the effect of the changes in haemoglobin combined with carbon monoxide and blood glucose concentration on the worker's headache and physical weakness.

(ii) Analyse the cause of the change in the speed of nerve impulse transmission on the worker's response to stimuli.

(b) Evaluate the effectiveness of the observed physiological responses in improving oxygen supply, maintaining body temperature and restoring blood glucose concentration.

ITEM 3

A wildlife centre in Queen Elizabeth National Park monitored a group of antelopes during a period of high environmental temperature. Some animals were exposed to a disease-causing bacterium, while the infection damaged the semicircular canals of the inner ear in a few animals. Dominant males also prevented weaker antelopes from feeding adequately during the day.

The affected antelopes showed excessive tiredness, unsteady movement and loss of body mass. Investigations produced the results shown in the table below.

Table: Changes recorded in the antelopes

Biological indicator	Before the stressful conditions	During the stressful conditions	Expected healthy level
Body temperature (°C)	38.4	41.2	38.0–39.0
Specific antibodies (arbitrary units)	4	37	Above 30 after infection
Incorrect steps while walking along a straight path	1	16	0–2
Feeding time per day for weaker antelopes (hours)	5.8	2.1	5–7

The antelopes developed the following physiological mechanisms:

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- increased panting and dilation of blood vessels beneath the skin;
- increased production of specific antibodies;
- increased use of visual and proprioceptive information to coordinate movement;

Task

(a) Analyse how the changes shown in the table contributed to excessive tiredness, control of the bacterial infection, unsteady movement and loss of body mass.

(b) Evaluate the effectiveness of each physiological mechanism in promoting the survival of the affected antelopes.

PART II
CONSTRUCT 4

ITEM 1

A woodland bird species originally occupied one continuous habitat in the Albertine region. Construction of a large settlement divided the habitat into two isolated areas. One area contained mainly hard seeds, while the other contained mostly soft fruits. Movement of birds between the areas became rare.

After many generations, birds in the two areas developed different beak shapes and courtship songs. Attempts to mate birds from the two populations produced very few fertile offspring. An invasive shrub also spread through both areas and replaced several native food plants.

The bird populations became small, and some local plant species failed to produce enough seeds.

Table: Findings from the separated woodland areas

Observation	Area A	Area B
Frequency of the broad-beak allele (%)	82	19
Successful mating with birds from the other area (%)	7	6
Native fruit-producing plants per hectare	63	21
Invasive shrubs per hectare	48	176

Conservation officers suggested the following measures:

- protect the remaining native food plants in both areas;
- remove the invasive shrubs gradually;
- introduce another rapidly spreading foreign shrub;

Task

(a) (i) Analyse how isolation and differences in food resources contributed to genetic divergence and speciation in the birds.

(ii) Explain how the invasive shrub affected the bird populations and reproduction of native plants.

(b) Assess the effectiveness of the proposed conservation measures in maintaining biodiversity and reducing the risk of extinction.

ITEM 2

A poultry farmer in Mukono District introduced newly hatched chicks into a brooder. Some chicks were exposed to a viral disease before vaccination. A pesticide used in the brooder interfered with the breakdown of acetylcholine at neuromuscular synapses. The farmer also provided salty drinking water and separated the chicks from the mother hen shortly after hatching.

The chicks later experienced repeated illness, uncontrolled muscle contractions, severe loss of body water and failure to follow the mother hen. The investigation findings are presented in Table below.

Table: Findings from different aspects of the chicks' condition

Aspect investigated	Healthy chicks	Affected chicks
Specific antibody concentration (arbitrary units)	68	14
Acetylcholine remaining in synaptic clefts (relative units)	1.2	7.6

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Urine produced per day ($cm^3\ kg^{-1}$ body mass)	24	8
Chicks following the mother hen (%)	91	23

The farmer proposed the following interventions:

- vaccinate the chicks against the viral disease;
- increase the pesticide concentration to stop the muscle contractions;
- replace the salty water with clean fresh water;

Task

(a) (i) Analyse how the changes in antibody concentration and acetylcholine contributed to repeated illness and uncontrolled muscle contractions.

(ii) Explain how the changes in urine production and following behaviour were related to water loss and separation from the mother hen.

(b) (i) Evaluate the effectiveness of vaccination and increased pesticide application in improving the health of the chicks.

(ii) Assess the effectiveness of providing fresh water and returning the chicks to the mother hen in correcting the remaining problems.

ITEM 3

A fish-breeding project near Lake Kyoga crossed tilapia that were heterozygous for fast growth and resistance to a bacterial disease with slow-growing, disease-susceptible tilapia. Fast growth, **G**, was dominant over slow growth, **g**, while disease resistance, **D**, was dominant over susceptibility, **d**.

Persistent pesticides from nearby farms entered the lake through surface runoff. The contaminants passed through feeding relationships in the lake and affected the survival of some offspring produced by the breeding project.

Fish harvests later declined, and consumers were warned against frequently eating mature tilapia from the affected area.

The offspring from the cross were:

- fast-growing and disease-resistant: 392;
- slow-growing and disease-susceptible: 401;
- fast-growing and disease-susceptible: 104;
- slow-growing and disease-resistant: 103.

Table: Findings from the lake food chain

Organism	Energy available ($kJ\ m^{-2}\ year^{-1}$)	Pesticide residue ($mg\ kg^{-1}$)
Algae	72,000	0.04
Zooplankton	7,100	0.48
Small fish	690	4.6
Mature tilapia	61	32.8

Project managers proposed the following interventions:

- breed from fast-growing, disease-resistant offspring;
- apply more of the persistent pesticide on farms near the lake;
- establish vegetation strips between farms and the lake;

Task

(a) (i) Analyse the inheritance pattern represented by the tilapia offspring and explain its importance in maintaining the fish population.

(ii) Analyse the changes in energy and pesticide residues along the food chain and explain how they affect the survival of genetically selected tilapia.

(b) Evaluate the effectiveness of the proposed interventions in improving the genetic quality of the fish stock, protecting the lake ecosystem and maintaining food security.

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